

CSCI 6840 Project 2

Handwritten Digit Classification Using Artificial Neural Networks in PyTorch

Objective:

Students will build, train, and evaluate an artificial neural network (ANN) for classifying handwritten digits using the MNIST dataset. You will implement the model using PyTorch, learn about optimization techniques, and tune the hyperparameters to improve performance.

Steps:

1. Dataset:

- Use the MNIST dataset, which contains 70,000 images of 28x28 grayscale digits (0-9). The dataset is available in the torchvision library.
- Split the dataset into training (60,000 images) and test (10,000 images) sets. Additionally, create a validation set of 10,000 images.

2. Model:

- Design a simple feedforward artificial neural network (ANN) with at least:
 - An input layer (784 input units, corresponding to 28x28 pixels).
 - At least two hidden layers with non-linear activation functions (e.g., ReLU).
 - An output layer with 10 units (for the 10 digit classes) using a softmax activation.

3. Loss Function & Optimizer:

- Use Cross-Entropy Loss for multi-class classification.
- Students can experiment with optimization algorithms such as SGD or Adam, and try different learning rates.

4. Training:

- Train the model for a minimum of 20 epochs, using batches of data.
- Plot the training and validation loss to observe overfitting or underfitting.
- Track accuracy as the main performance metric.

5. Implementation in PyTorch:

- Students will implement the model architecture, training loop, and evaluation metrics in PyTorch.

6. Evaluation:

- Evaluate the model on the test set and report classification accuracy.
- Optional: use techniques like learning rate scheduling or early stopping.

7. Submission Requirements:

- Well-documented code in Python (.py file).
- A report (6+ pages) explaining the model design, hyperparameter tuning, training progress (with plots of loss and accuracy), and test set performance.
- Include graphs showing the training loss, validation loss, and accuracy over epochs.

Deliverables:

- Python code in .py file.
- A report discussing model design, hyperparameter tuning, and results.
- Presentation slides. In your presentation, you are required to explain your codes.